

Aerodynamics and Design of Aerospace Flight Vehicle

Course Code: U02M12525

Final Project:
CFD Simulations Report

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Table of Contents:

I-Objectives

II-Geometry specifications

III-Operating conditions

IV-Software utilized

V-Simulation results

VI-Discussion

VII-Limitations and uncertainty

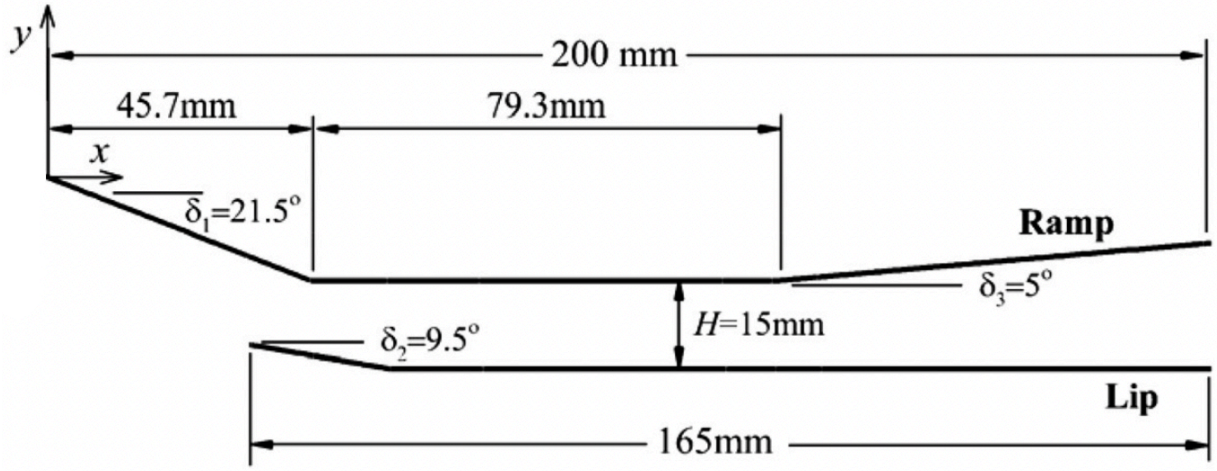
VIII-Conclusion

I-Objectives:

- Get familiar with the usage of CFD software.
- Replicate the configuration of the supersonic inlet isolator from the reference paper.
- Simulate it under the appropriate operating conditions.
- Compare the obtained results with the previous findings.
- Analyze the flow pattern further.

II-Geometry specifications:

The inlet design is adopted from the reference paper. It consists of three segments: a compression ramp of 45.7mm with a convergence angle δ1=21.5 deg, followed by a 79.3 mm constant area isolator, and then a 75 mm expansion ramp with a divergence angle δ3=5 deg. The height of the isolator, H, is 15 mm. The isolator lip, with an inclination angle δ2=9.5 deg, starts (axially) at 35 mm downstream of the ramp starting point and has a total length of 165 mm.



III-Operating conditions:

The flowing fluid is air, considered an ideal nonreactive dry gas with constant specific heat and thermal conductivity. A finite volume technique is used to solve two-dimensional compressible Reynolds-averaged Navier-Stokes (RANS) equations. Turbulent properties are modeled using a two-equation k–ω shear stress transport of Menter, which formulates the modified turbulent viscosity to account for the transport effects of the principal turbulent shear stress.

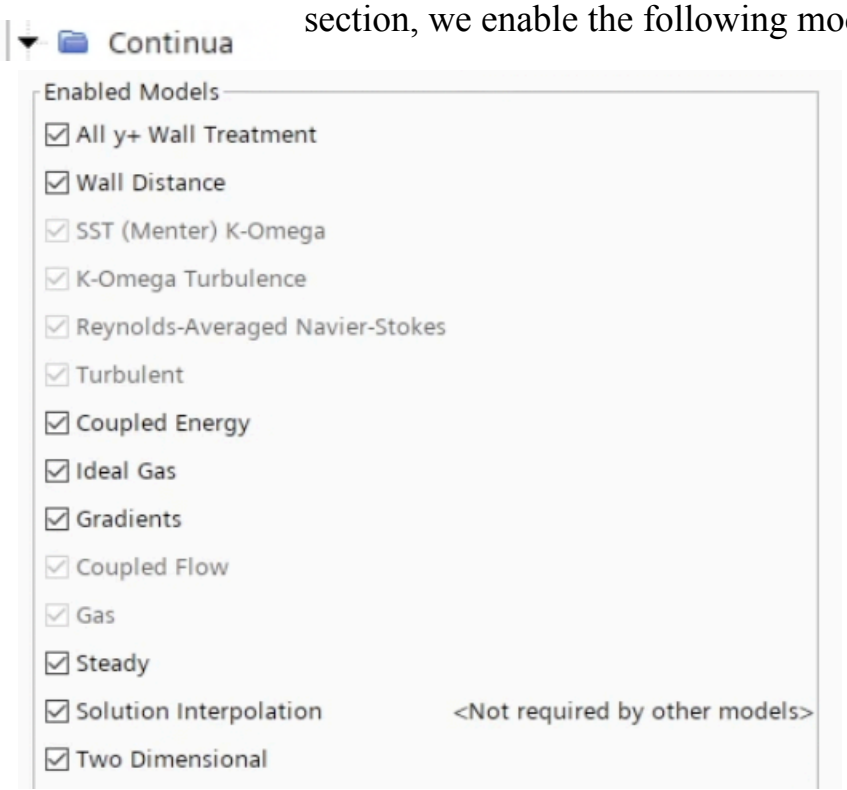
- Mach numbers: 2, 2.41, 3
- Angle of attack $\alpha = -10$ degrees
- Total pressure P0 = 5.4 bar
- Total temperature T0 = 305 K

The solid surfaces of the inlet model are considered a no-slip, impermeable, adiabatic, and stationary wall boundary conditions.

IV-Software Utilized:

The CFD simulations were performed using Star CCM+. Star CCM+ is a widely used computational fluid dynamics software known for its robustness in simulating complex fluid flow phenomena. It incorporates advanced models and algorithms, enabling accurate representation of aerodynamic behaviors. The software's capabilities facilitated the analysis of flow patterns, pressure distributions, and other relevant parameters essential for the objectives outlined in this study.

To set up the fluid physics for our simulation, we first define the properties of the working fluid as follows:

Under the 

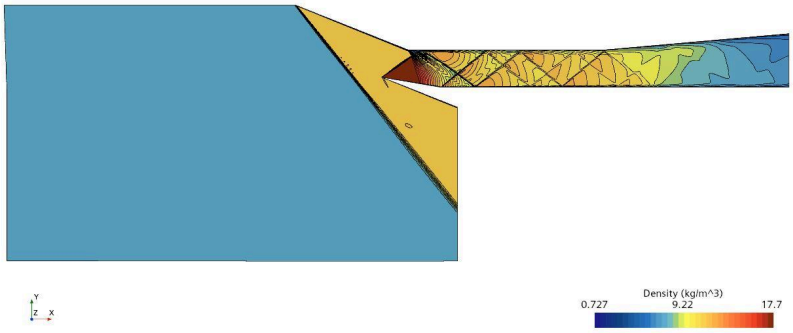
section, we enable the following modes:

The criterion for assessing convergence was based on visual assessment. The computations were stopped when the residuals became ever so slightly varying, usually after a 1000 iteration.

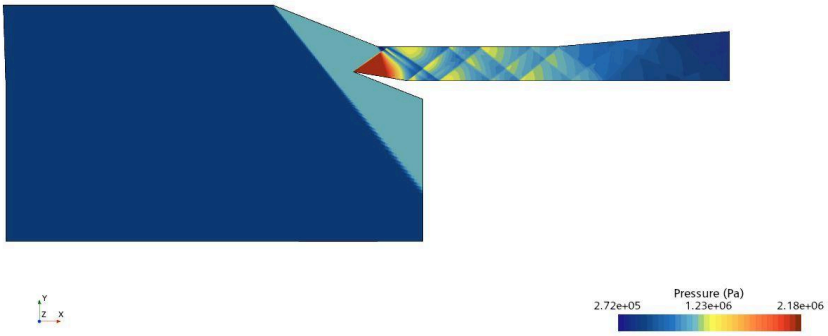
V-Simulation results:

1-Mach=2:

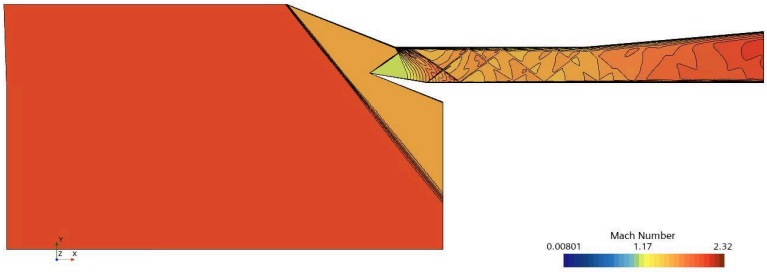
- **Density:**



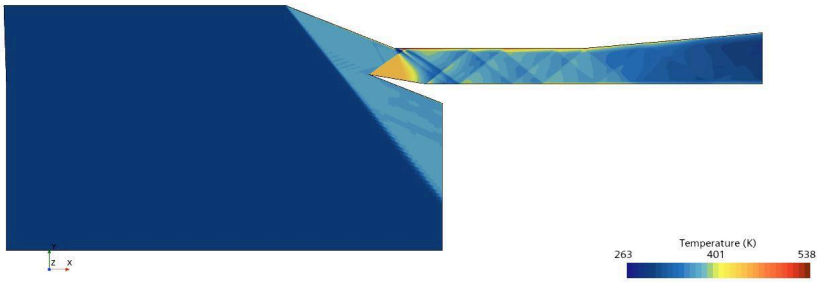
- **Pressure:**



- **Mach number:**

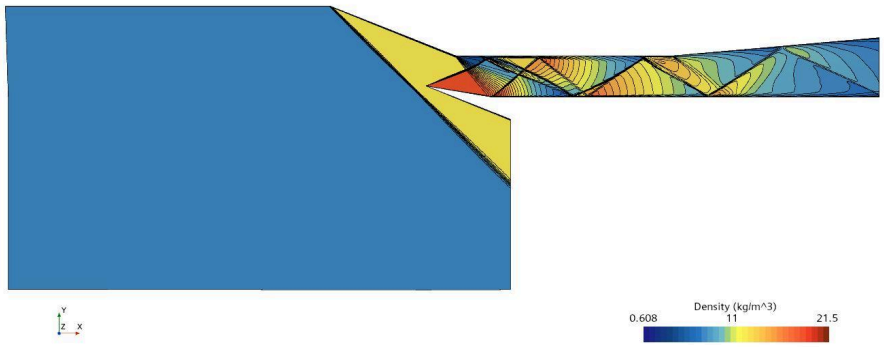


- **Temperature:**

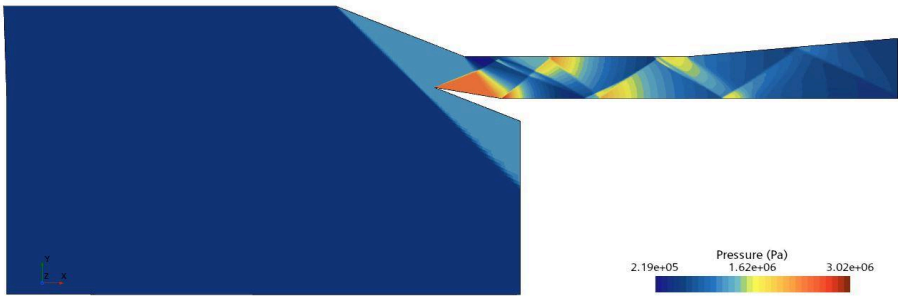


2-Mach=2.41:

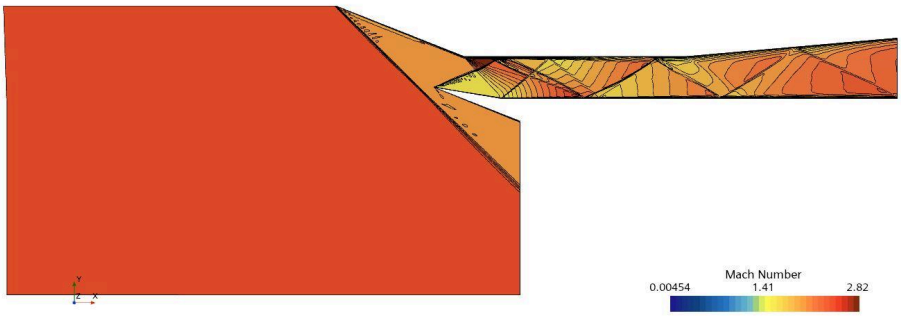
- **Density:**



- **Pressure:**



- **Mach number:**

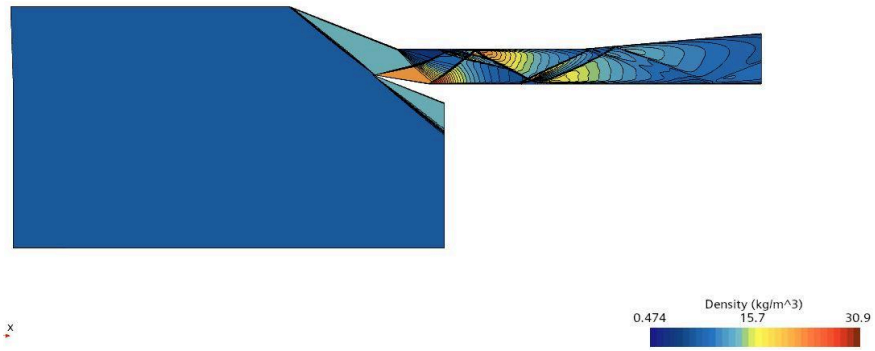


- **Temperature:**



3-Mach=3:

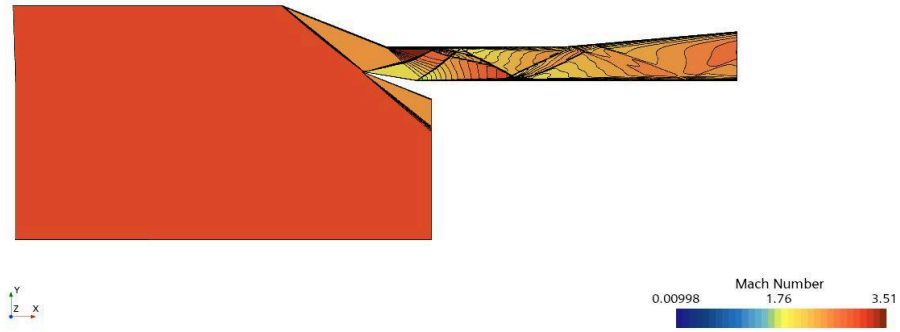
- **Density:**



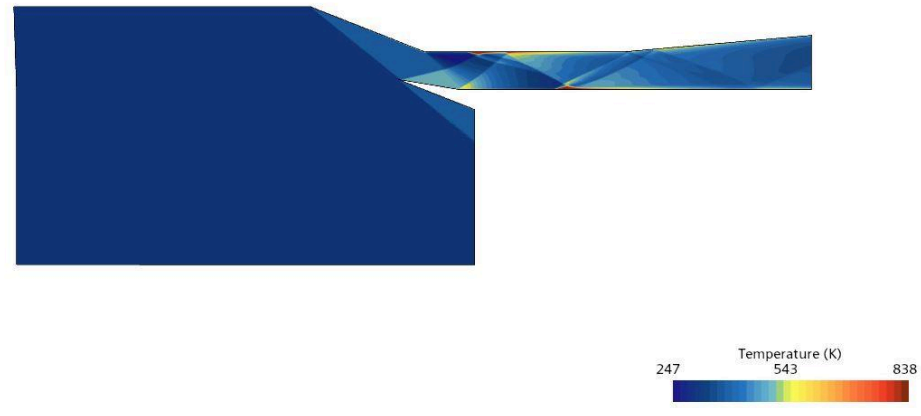
- **Pressure:**



- **Mach number:**

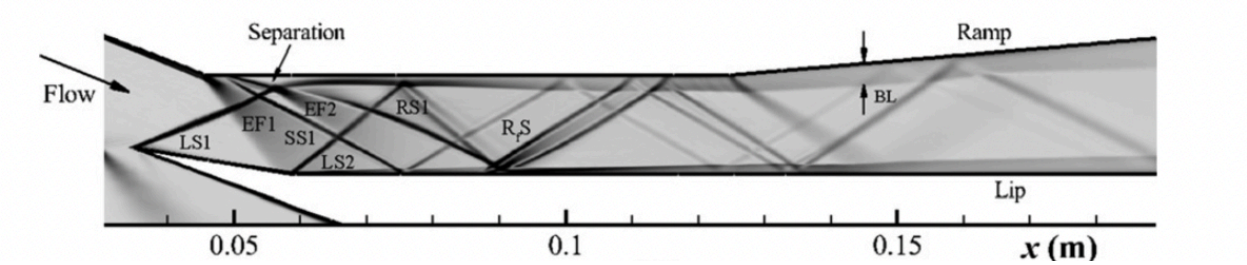


- **Temperature:**

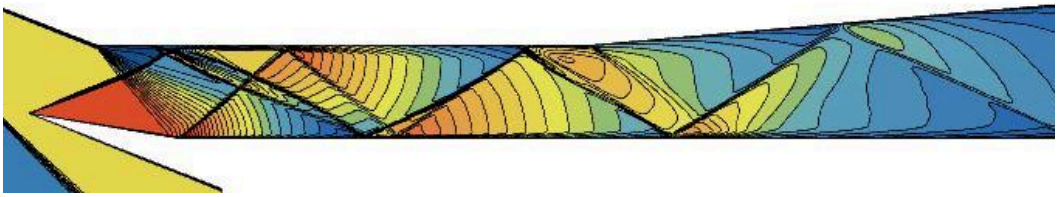


VI-Discussion:

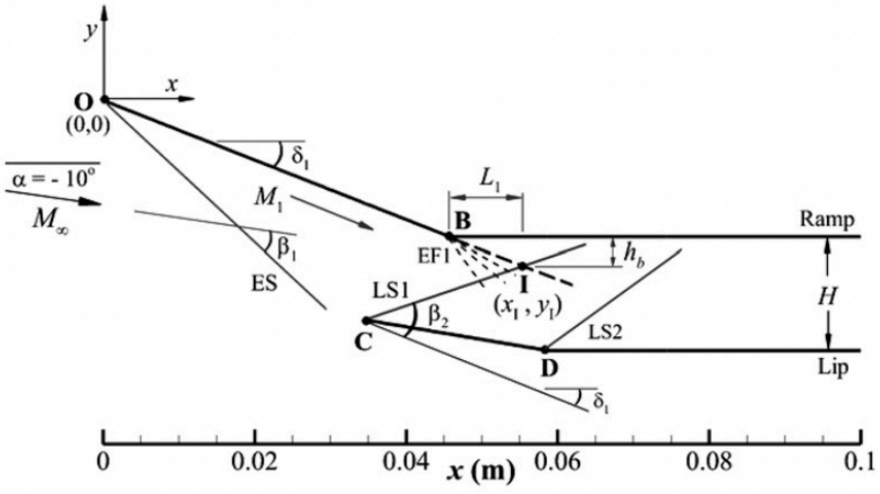
The obtained results can be validated by comparison with the available experimental data. Fig. 4(b) from the paper shows the experimental density gradient at a Mach=2.41.



A similar flow structure was developed in the conducted simulation, as can be seen below:

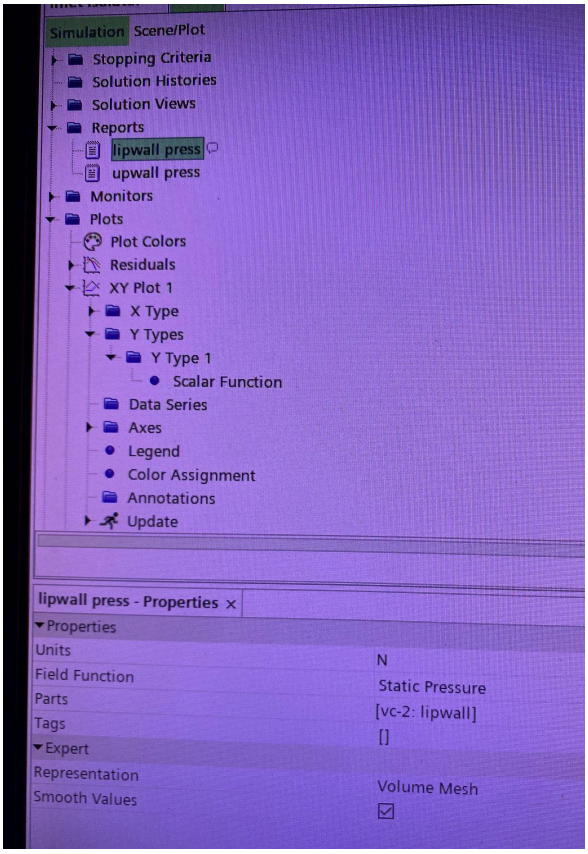


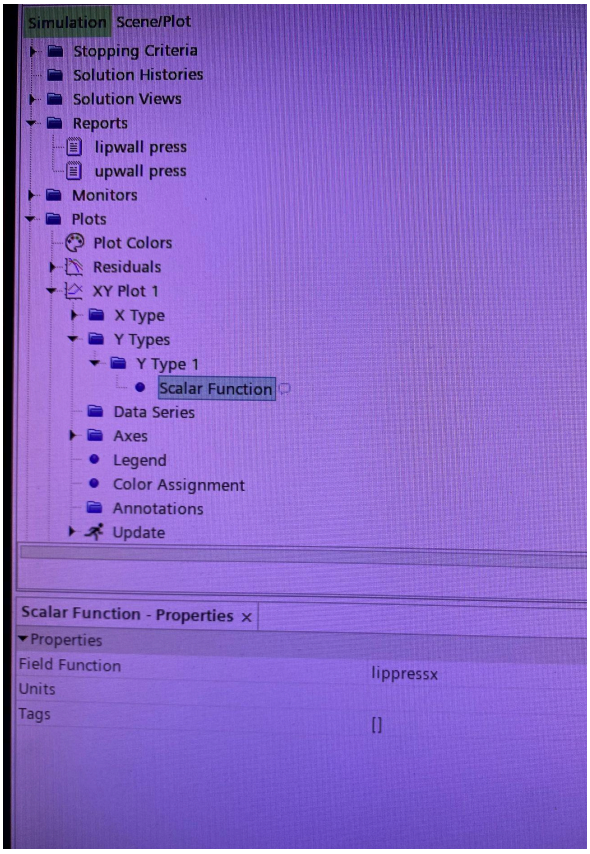
The analysis of the flow field around the inlet is primarily based on the theory of inviscid flow of oblique shock waves. In this case, the flow will first experience a compression ramp, which generates an external oblique shock at point O. For a given freestream Mach number M∞ (at a=−10 deg), the shock wave angle β1 and downstream Mach number M1 are calculated from the inviscid θ–β–M relation. Then, this flow with Mach number M1 is compressed once again at point C through a lip shock. The shock wave angle β2 is theoretically determined using the same θ–β–M relation.

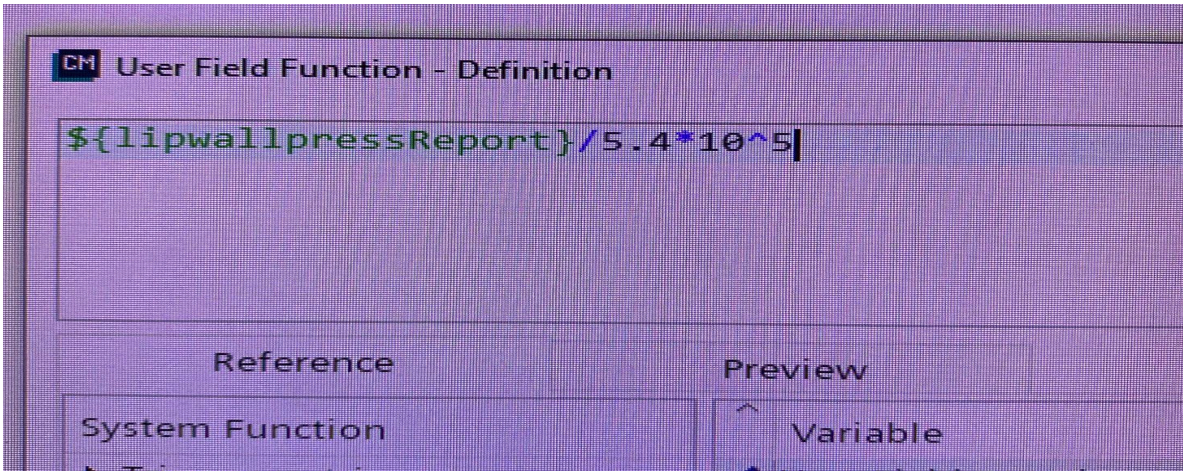


VII-Limitations and uncertainty:

Several challenges arose while attempting to generate the P/P0 graph for the upper ramp and the lower lip, despite consulting online documentation. Initially, a syntax error hindered progress, which was resolved after debugging (5.4*10^5 had to be changed to 5.4e5). However, despite these efforts, the software failed to display any results upon subsequent runs. This issue stems from a combination of limited familiarity with the software and CFD simulations overall, in addition to the time constraints to complete the tasks.







VIII-Conclusion:

In summary, the CFD simulation of the supersonic inlet isolator has been a valuable learning experience, providing hands-on insight into its aerodynamic behavior. Despite being the first attempt at CFD, the simulation offered a real-life visualization of the theoretical concepts studied. It revealed crucial flow characteristics such as shockwave formation and pressure distribution, bridging the gap between theory and practical application. While there's room for improvement and validation against experimental data, this project has laid a solid foundation and sparked further interest in future studies in CFD simulation.

The referenced paper throughout this study is:

Khan, M. R. A., and Hasan, A. B. M. T. (February 9, 2021). "Design and Evaluation of Generic Bump for Flow Control in a Supersonic Inlet Isolator." ASME. *J. Fluids Eng*. May 2021; 143(5): 051207.